Ekvin Saleh

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Education

Delft University of Technology — Delft, Netherlands
Master in Applied Physics, Track: Energy Materials — *Sep 2025 – Present*

University of Twente — Enschede, Netherlands
Pre-Master Applied Physics — *2024*

University of Twente — Enschede, Netherlands
Bachelor in Electrical Engineering — *2021 – 2024*

- Thesis: *Characterization of Electric Double-Layer Capacitors Using Electrochemical Impedance Spectroscopy*
- Grade: 8.5

Experience

Embedded Software Engineer — May 2025 – Sep 2025
Royal Pas Reform — *Zeddam, Netherlands*

- Maintained and extended a C++ state-machine tester for PLC-controlled hatchery control boxes, using CSV-defined profiles on Debian systems.
- Implemented automated UPS and regional power-configuration checks for Canadian-bound systems.
- Debugged and maintained the company's Python fork of *websockify*, improving its logging.
- Built custom lightweight Docker images from scratch for MariaDB, Apache/PHP, and *websockify*, served from an internal registry for remote hatchery machines.
- Investigated hardware test warnings by comparing schematics with MOSFET/transistor datasheets and adjusting tester logic.

Cook — Dec 2022 – Apr 2025
Fabels — *Enschede, Netherlands*

- Owned daily mise en place across all stations in a high-volume bistro kitchen.
- Worked peak dinner services plating 200+ covers per shift, maintaining consistency under sustained pressure.
- Closed kitchen independently: deep-cleaning stations, inventory checks, and next-day prep.

Teacher Assistant — Sep 2022 – Apr 2023
University of Twente — *Enschede, Netherlands*

- Supervised and graded lab sessions on amplifier designs using BJTs and MOSFETs.
- Supported first-year students with MATLAB and lab equipment.

Technical Skills

Languages: Python, C++, SQL, MATLAB (Simulink), Mathematica

Scientific Computing: NumPy, SciPy, CuPy, Matplotlib, pytest, vectorized simulation, Monte Carlo methods

Developer Tools: Git, SVN, Docker, Apache, Bash, Makefiles, LaTeX

Hardware & Simulation: LTspice, VHDL, Altium

Fabrication & Characterization: 3D printing, laser cutting, cleanroom project exposure (magnetron sputtering, lithography workflow, SEM)

Visualization: Autodesk Maya, VTK, VisPY

Publications

E. Saleh, et al., "EIS-Based State of Charge Characterization of Electric Double-Layer Capacitors," *IEEE ECCE*, 2025. [View Publication]